NCL Resource Guide

Season Overview

While there is an access period for the Gymnasium and the games, you don't have to work on them for the entire duration. You can work on the challenges at any time during the access period at your pace.

- **Gymnasium:** A practice environment with guides of past challenges to help players prepare for the competition.
- **Practice Game:** A week-long event that gives practice solving challenges without guides.
- **Individual Game:** A weekend-long competitive event where individual players will attempt to solve as many challenges as they can.
- **Team Game:** A weekend-long competitive event. Players will compete in teams of up to 7 and work to solve as many challenges as they can. You can also participate as a team of 1 if you would like to compete solo.

General Resources

- Season Schedule
- Registration Link
- Trove
- Rules of Conduct
- Gymnasium
- YouTube
- Discord
- More Season Info

TOPIC OVERVIEW

Open Source Intelligence:
Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

Log Analysis:
Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

Scanning & Reconnaissance:
Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

Cryptography:
Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plain text.

Network Traffic Analysis:
Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

Web Application Exploitation:
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

Password Cracking:
Identify types of password hashes and apply various techniques to efficiently determine plain text passwords.

Forensics:
Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

Enumeration & Exploitation:
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

Suggested Practice Schedule

Before you start diving into your NCL practice, please keep in mind that NCL contains challenges that span a wide range of topics and skill levels, and you are not expected to know or learn everything covered here during the course of a single NCL season. The suggested schedule is here as a guide to help you plan out a practice schedule that covers all of the topics.

You may want to spend more time on a specific topic or skip some topics entirely, and that is perfectly fine. During the games, you will have the ability to choose the topics you would like to work on. The goal is not to have you master all of the skills in a single season, it is to have you learn and discover something new each season.

Week 1: Fundamentals and Open Source Intelligence

Learning Resources
Computer Fundamentals for Cybersecurity
Open Source Intelligence Basics

Tools
Maxmind (GeoIP Lookup)
Archive.org (Web Archive)
OSINT Framework

Gym Challenge Guides
Open Source Intelligence - Meta
Open Source Intelligence - Lookup
Open Source Intelligence - Threat Intel
Open Source Intelligence - HTTP Headers
Open Source Intelligence - SSL
Open Source Intelligence - Barcode
Open Source Intelligence - WHOIS
Open Source Intelligence - PGP Lookup

Week 2: Cryptography & Password Cracking

Learning Resources
How Does Binary Work?
Decrypting an RSA message
Password Cracking with Hashcat
Cracking Password Hashes
Making Custom Wordlists for Password Cracking
Breaking Crypto with Known Plaintext Attacks
Cracking Crypto with CyberChef

Tools
Cyberchef
Rumkin
Quipquip
Hashcat
John the Ripper
Ophcrack
hashID

Gym Challenge Guides
Cryptography - Number Bases
Cryptography - Shift
Cryptography - @bash
Cryptography - Beep
Cryptography - Fencing
Cryptography - French
Cryptography - Strings
Cryptography - RSA
Password Cracking - Rockyou
Password Cracking - Mask
Password Cracking - Pokemon
Password Cracking - Windows
Password Cracking - Law & Order
Password Cracking - PDF
Password Cracking - Kali Linux

Week 3: Log Analysis

Learning Resources
Linux Command Line
Linux Command Line Part 2
Log Analysis 1
Log Analysis 2
Analyzing an SQL Database
Writing Code for Log Analysis

Tools
SQLite Viewer
Cut
Sort
Uniq
Grep
Wc
OverTheWire: Bandit

Gym Challenge Guides
Log Analysis - SSH
Log Analysis - Nginx
Log Analysis - History
Log Analysis - Squid
Log Analysis - Payments
Log Analysis - VSFTPD
Log Analysis - Login
Log Analysis - Custom File Format

Week 4: Network Traffic Analysis

Learning Resources
Understanding TCP
Pinging All Devices - Please Echo Back
ICMP
DNS
What is Telnet
Analyzing FTP Traffic
Reverse Engineering A Network Protocol
Cracking WiFi Passwords
Decrypting SSL Traffic with an SSL Key

Tools
Wireshark
Aircrack-ng

Gym Challenge Guides
Network Traffic Analysis - DNS
Network Traffic Analysis - FTP
Network Traffic Analysis - HTTP
Network Traffic Analysis - Telnet
Network Traffic Analysis - Pandora
Wireless Access Exploitation - Packet Dissection
Wireless Access Exploitation - Decrypt
Wireless Access Exploitation - CAN Bus

Week 5: Forensics & Scanning

Learning Resources
Forensics w/ Binwalk
Using Git to Find Files (and Secrets)
Nmap Scan Demo

Tools
Binwalk
Nmap
Dirbuster

Gym Challenge Guides
Forensics - Version Control
Forensics - File Carving
Forensics - PDF
Forensics - Magic Bytes
Forensics - Docter
Forensics - The Book
Scanning - NMAP
Scanning - Metadata
Scanning - Net Track
Scanning - Git

Week 6: Web Application Exploitation

Learning Resources
Hacking with Chrome Dev Tools
SQL Basics
Exploiting Untrusted Web Data
Manipulating Web Cookies
Man in the Middle with Burp Suite

Tools
Chrome DevTools
Burp Suite Community Edition

Gym Challenge Guides
Web Application Exploitation - egov
Web Application Exploitation - Metro Lottery
Web Application Exploitation - Metro Clinic
Web Application Exploitation - Never Winter Break
Web Application Exploitation - Leek

Week 7: Enumeration & Exploitation

Learning Resources

Code Basics
Code Basics 2
101 on Code Execution Basics
Code Reverse Engineering Basics
Ghidra: The NSA Reverse Engineering Tool
Exploiting vsftpd
Revealing Python with Uncompyle

Tools

Ghidra
Uncompyle
Metasploit

Gym Challenge Guides

Enumeration & Exploitation - Python 1
Enumeration & Exploitation - Python 2
Enumeration & Exploitation - Python 3
Enumeration & Exploitation - Binary 1
Enumeration & Exploitation - Binary 2
Enumeration & Exploitation - Timebomb
Enumeration & Exploitation - Bytes